

..... Massive Crowd Management in Urban Pedestrian Zones

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Abstract—This paper presents a bi-level optimization framework for crowd management in large pedestrian zones, integrating discrete directional constraints with anisotropic geometric guidance. The lower level employs a Hughes-type macroscopic crowd model enhanced with constrained Hamilton-Jacobi-Bellman (HJB) equations for one-way, bidirectional, and closed channel states, and then uses metric tensor fields to represent soft channel-alignment guidance within the feasible direction set. The upper level optimizes channel direction configurations and guidance intensities using a Structure-Aware Hybrid Block Optimization (SA-HBO) algorithm. Three raw physical metrics are introduced for efficiency, safety, and load balancing, and are further converted into dimensionless indicators for cross-scheme comparison and scalarization. The proposed method combines surrogate pre-screening, full simulation confirmation, and block-coordinate updates to search the mixed discrete-continuous control space. Numerical experiments demonstrate the effectiveness of the proposed approach.

Index Terms—Crowd dynamics, macroscopic modeling, channel optimization, Hamilton-Jacobi-Bellman equation, bi-level optimization.

I. INTRODUCTION

LARGE-SCALE crowd gathering risks not only threaten individual safety but may also escalate into stampede incidents that disrupt social order and public security; the effectiveness of crowd management directly tests urban governance capacity and the modernization level of social safety governance systems [1], [2]. With the increasing frequency of high-density pedestrian activities such as holiday tourism, urban public events, commercial district promotions, New Year celebrations, and major sports events, core urban areas must accommodate pedestrian demands far exceeding daily levels within short time windows, posing enormous challenges for on-site organization, passage management, risk early warning, and emergency dispatch [3], [4], [5]. Against this backdrop, scientific modeling, numerical simulation, and optimization methods for formulating and evaluating crowd control strategies—improving the efficiency of large-scale event organization and reducing local congestion risks—have become critical scientific problems demanding urgent breakthroughs in social safety and urban governance [6], [7].

Crowds in large-scale events constitute typical complex systems. Numerous individuals with heterogeneous desired speeds, destinations, and route preferences interact through multi-scale, nonlinear mechanisms in confined spaces, generating spatiotemporal patterns at the collective level [8], [4]. Macroscopically, these patterns manifest as destination-oriented flow, passage convergence, bottleneck queuing, exit competition, and directional responses to control regulations

[9], [10]; microscopically, individuals are influenced by surrounding density, visible paths, obstacle boundaries, and guidance facilities [11]. For open scenic areas, crowd behavior additionally exhibits pronounced phase dependence: visitors may first move toward the viewing platform, then tour along the main viewing direction, and subsequently choose different exit passages based on preferences [12]. These characteristics of multi-stage, multi-route coexistence with shared congestion mean that traditional static capacity analysis or single shortest-path models cannot adequately support refined management decisions.

Among various urban crowd gathering scenarios, open scenic areas, waterfront viewing platforms, and historic commercial districts are highly representative. Unlike enclosed venues such as stadiums and subway stations with well-defined boundaries [13], [14], such areas consist of open spaces, main tourist corridors, multiple lateral connecting passages, and return flow lines; visitors typically undergo a continuous behavioral chain of “entry–touring along the main corridor–choosing passages to leave–returning” [15], with crowd movement governed jointly by spatial geometry, control regulations, and historical behavioral preferences [16], [15]. Taking the Shanghai Nanjing East Road–Bund area as a concrete example, visitors approach the Bund from Nanjing East Road, ascend to the viewing platform via multiple stepped passages, tour along the waterfront, then exit through several southern passages and return along lower streets. This scenario can be abstracted as the typical “waterfront platform–multi-stepped passage” crowd flow scenario. In such scenarios, management difficulties manifest in three aspects: first, multiple passages simultaneously serve entry, exit, and diversion functions, and local direction settings can alter overall flow organization; second, distinct phase-dependent behavioral transitions exist between the main tourist corridor and lateral passages, making single origin–destination models insufficient; third, local congestion and global route choice form mutual feedback loops, where control changes in one passage may trigger load redistribution in adjacent areas. Fig. 1 shows the large-scale crowd at the Shanghai Nanjing East Road–Bund scenario during a holiday period.

Existing large-scale event security management practices typically integrate system dynamics, system coupling, and system safety theory, combined with high-risk zone delineation and historical crowd behavior patterns, to formulate a series of control measures [17], [18]. These measures include deploying security personnel at critical locations, erecting temporary barriers for isolation and diversion, implementing one-way passage on key channels, and applying batch release in core areas [19], [20]. For instance, during major holidays, the Shanghai Bund area adopts batch release, passage direction regulation, south-entry north-exit routing, and reduction of



Fig. 1. Large-scale crowd at the Shanghai Nanjing East Road–Bund scenario during a holiday period.

head-on conflicts [21]; similarly, Nanluoguxiang main street in Beijing implements one-way passage measures (south entry, north exit) during peak periods [22], consistent in control mechanism with the Shanghai Bund approach.

However, current management measures for urban high-pedestrian-flow areas still rely considerably on manual experience and static contingency plans, with deficiencies remaining in scientific evaluation and refined optimization [23]. First, many control schemes are formulated based on historical experience, lacking quantitative characterization of the impact of different passage direction configurations, geometric guidance intensities, and phase-dependent behavioral preferences. Second, existing methods tend to focus on single-point statistics or local density monitoring, making it difficult to describe the dynamic coupling among “passage direction rules–local optimal directions–collective density evolution–exit load distribution” [24]. Third, in open scenic area scenarios with multiple entrances, passages, and stages, neglecting visitor route preferences and phase transition behaviors can easily lead to misjudgment of passage loads and high-risk area locations. Finally, many existing simulation or optimization methods treat the crowd model as a black-box evaluator, failing to exploit the structural information among passage geometry, one-way rules, and multi-route subpopulations [25], resulting in insufficient efficiency in control strategy search [26].

To address the above issues, this paper proposes a macroscopic crowd modeling and control optimization method incorporating passage constraints, geometric guidance, and multi-stage route behavior, targeting the “waterfront platform–multi-stepped passage” open scenic area scenario. At the modeling level, this paper constructs a Bellman–conservation law coupled crowd continuum model: different stage–route subpopulations possess independent potential functions with optimal directions computed via the discrete Bellman equation, while sharing the speed–density relationship determined by total density; one-way, bidirectional, and closed passage states are formulated as constraints on the local admissible direction set, and an anisotropic metric tensor characterizes the cost differences between axial and lateral movement, simulating

pedestrian pre-alignment and streamline reorganization near passages. At the optimization level, this paper distinguishes “controllable management variables” (passage direction configuration and geometric guidance intensity) from “exogenous behavioral parameters” (visitor route diversion preferences identified from historical data). Based on this boundary, total travel time, high-density exposure time, and channel cumulative flow variance are adopted as efficiency, safety, and load balancing indicators, forming a scalarized optimization problem under fixed weights. A simplified instance of the Structure-Aware Hybrid Block Optimization framework (SA-HBO) is designed to solve this problem, exploiting the discrete neighborhood structure of passage direction variables and the continuous nature of geometric guidance variables through surrogate pre-screening, full simulation confirmation, and acceptance/rejection mechanisms.

The main contributions of this paper are as follows.

- 1) A “waterfront platform–multi-stepped passage” crowd flow modeling framework for open scenic areas is proposed, representing visitor behavior as a multi-stage, multi-route continuum flow process, where different subpopulations possess independent potential functions while sharing collective congestion feedback.
- 2) A Bellman–conservation law coupled model integrating passage constraints and geometric guidance is constructed. The admissible direction set first encodes one-way, bidirectional, and closed channel states, while an anisotropic metric tensor simulates pre-alignment, smooth convergence, and streamline reorganization near passage entrances.
- 3) An evaluation framework distinguishing mechanism verification from strategy optimization is established, categorizing indicators into behavioral mechanism indicators (direction consistency, approach angle distribution, channel capture domain) and management optimization indicators (total travel time, high-density exposure time, channel flow variance).
- 4) A structure-aware optimization framework for controllable passage configuration is designed, jointly optimizing passage direction configuration and geometric guidance intensity under fixed historically identified preference parameters through discrete neighborhood search, continuous local search, and surrogate pre-screening with acceptance/rejection mechanisms.

II. RELATED WORK

A. Massive Crowd Management and Control

Crowd management in urban centers has been extensively studied from control-theoretic and optimization perspectives, aiming to enhance safety and efficiency through boundary regulation, facility layout, and intelligent decision-making [27], [28].

In control-theoretic approaches, Zhong et al. [29] designed boundary control strategies for urban traffic flow networks to ensure convergence to uncongested states under disturbances, with stability analysis based on Lyapunov methods [30]. For one-dimensional crowd dynamics, Wadoo and Kachroo [31]

employed advection and diffusion control to prevent blocking and shock waves during evacuation. Addressing multi-directional disturbance propagation, Qin [32] proposed a unit sliding mode controller based on integral barrier Lyapunov functions to ensure boundedness of internal state variables, demonstrating strong disturbance rejection capabilities. For heterogeneous corridor regulation, Zhu et al. [33] combined policy learning with neural networks to learn optimal feedback controllers that adjust inflow rates and free-flow speeds, achieving balanced density distribution and reduced congestion.

Physical infrastructure regulation has also proven effective. Studies have shown that strategically placed obstacles can significantly improve outflow, velocity, and local density in merging areas [23], [34]. Yang et al. [35], [36] employed Model Predictive Control (MPC) to optimize barrier lengths at subway bottlenecks, later extending control variables to entrance limiters, gates, and escalator directions for dynamic, real-time crowd regulation. For evacuation guidance, Carmona and Paricio-Garcia [37] proposed dynamic exit-choice recommendations using multinomial Logit models. However, their approach relies on discrete speed instructions (e.g., “stop,” “walk,” “run”) transmitted via electronic wristbands [38], which poses implementation challenges in open urban scenarios.

Recently, intelligent optimization and deep reinforcement learning have emerged as promising alternatives. Liao et al. [39] formulated fence layout as a subset selection problem, while subsequent work [26] proposed a dynamic crowd inflow control system using LSTM networks to capture temporal features and Proximal Policy Optimization (PPO) [40] for real-time entrance flow regulation. Differential Evolution (DE) [41] has also been applied to optimize guardrail layouts and flow guidance in real-world scenarios such as Chengdu East Railway Station, using average travel time and crowd pressure as multi-objective criteria [42].

B. Macroscopic Crowd Modeling

Macroscopic crowd models treat pedestrian flows as continuous media, offering computational efficiency and analytical tractability compared to microscopic approaches. Henderson [43] first applied fluid dynamics to pedestrian traffic in the early 1970s, comparing crowd motion to the behavior of gas or fluid particles.

Hughes [44] pioneered the first-order continuum theory for pedestrian flow, introducing the classical model in which pedestrian walking speed is governed by local density through a fundamental diagram, while movement direction follows the steepest-descent path of a potential function representing the remaining travel time to exits. This framework elegantly captures the interplay between congestion and route choice. Ling et al. [45] subsequently generalized the Hughes formulation and developed an efficient numerical solution combining a weighted essentially non-oscillatory (WENO) scheme for the conservation law with a fast sweeping method for the eikonal equation.

Several extensions have enriched the descriptive capability of macroscopic models. Nonlocal flux models were proposed

by Colombo et al. [46], replacing pointwise density with a neighborhood-averaged quantity in the velocity function to account for anticipatory behavior. This direction was further extended to incorporate anisotropic perception fields and boundary effects from walls and obstacles, with high-resolution numerical schemes and well-posedness results suitable for realistic architectural configurations [47]. To enforce the hard incompressibility constraint—that pedestrian density cannot exceed a physically admissible maximum—Maury et al. [48] introduced the projection model, which formulates the actual velocity as a least-squares projection of the desired velocity onto the set of feasible (non-overlapping) configurations. This framework was later recast as a Wasserstein gradient flow with density constraints [49], using the Jordan–Kinderlehrer–Otto (JKO) scheme [50] to discretize the motion while guaranteeing that density respects an upper bound, making it well-suited for high-density congestion scenarios.

To overcome non-physical behaviors such as supersonic information propagation that may arise in first-order models, Aw and Rascle [51] introduced a second-order model based on a generalized momentum conservation law that preserves anisotropy. The Aw–Rascle (AR) framework was subsequently adapted to pedestrian flows [52], where the congestion pressure term successfully reproduces characteristic phenomena including stop-and-go waves and spontaneous lane formation.

Beyond the classical Hughes-type framework, extensions have incorporated control and guidance mechanisms relevant to crowd management. Anisotropic formulations using metric tensors have been developed to represent channel geometries and directional preferences [53], [54], allowing for continuous geometric guidance without explicit prohibition. Constrained Hamilton–Jacobi–Bellman (HJB) formulations have been employed to model strict directional constraints such as one-way corridors, with the Bellman optimality principle providing a natural framework for unifying anisotropic geometric guidance with discrete directional constraints.

Despite these advances, existing approaches often treat control parameters as fixed or manually tuned. The bi-level optimization of discrete channel direction configurations combined with continuous guidance intensities remains underexplored, particularly for large-scale urban pedestrian zones with multiple behavioral stages and heterogeneous route choices.

III. MACROSCOPIC CROWD DYNAMICS MODEL

A. Hughes Model Foundation

We consider a bounded two-dimensional domain $\Omega \subset \mathbb{R}^2$ representing the pedestrian walkway. Let $\rho(x, t)$ denote the crowd density at position x and time t , $\phi(x, t)$ the minimal remaining time to reach an exit, and $\mathbf{v}(x, t)$ the mean velocity field. The continuity equation governing mass conservation reads:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (1)$$

In the classical Hughes model [44], pedestrians move in the direction of steepest descent of the potential function:

$$\mathbf{v} = f(\rho) \frac{-\nabla \phi}{|\nabla \phi|}, \quad (2)$$

where $f(\rho)$ is the speed-density relationship. We adopt the Greenshields form [55]:

$$f(\rho) = v_{\max} \left(1 - \frac{\rho}{\rho_{\max}}\right), \quad (3)$$

with $v_{\max}$ the free-flow speed and $\rho_{\max}$ the jam density.

The potential function satisfies the eikonal equation:

$$|\nabla\phi| = \frac{1}{f(\rho)}. \quad (4)$$

While this model captures basic congestion effects, it assumes that all directions are feasible and equally weighted. It therefore cannot represent one-way rules, temporary closures, or channel-induced geometric preferences. We address these limitations by first introducing HJB admissible-direction constraints for hard channel rules and then introducing the metric tensor to encode soft geometric guidance within the feasible direction set.

B. Channel Control via Constrained HJB

Many crowd management measures directly restrict available movement directions rather than merely adjusting traversal costs. Examples include one-way corridors, bidirectional passages, and temporary closures. Such rules are hard feasibility constraints on the local action set and are naturally modeled via constrained optimal control.

Let $U(x)$ denote the set of admissible unit directions at position x . The velocity satisfies:

$$\mathbf{v} = f(\rho)\mathbf{u}, \quad \mathbf{u}(x) \in U(x), \quad |\mathbf{u}| = 1. \quad (5)$$

By the Bellman optimality principle, the potential function satisfies the constrained HJB equation:

$$\max_{\mathbf{u} \in U(x)} \{-f(\rho)\mathbf{u} \cdot \nabla\phi\} = 1. \quad (6)$$

When $U(x)$ is the full unit circle, (6) reduces to the classical Hughes model. For restricted control sets, various channel rules can be expressed naturally. For channel c with region Ω_c and reference tangent $\tau_c(x)$, define

$$s_c \in \{-1, 0, +1, \emptyset\}, \quad (7)$$

where $+1$, -1 , 0 , and $\emptyset$ denote forward one-way, reverse one-way, bidirectional, and closed states, respectively. The corresponding admissible set is

$$U_c(x; s_c) = \begin{cases} \{\tau_c(x)\}, & s_c = +1, \\ \{-\tau_c(x)\}, & s_c = -1, \\ \{\tau_c(x), -\tau_c(x)\}, & s_c = 0, \\ \emptyset, & s_c = \emptyset, \end{cases} \quad x \in \Omega_c. \quad (8)$$

For a closed channel, the continuous maximum over an empty set is not used as an ordinary HJB state. Numerically, this state is handled as no feasible control in the channel region, equivalently $\phi(x) = +\infty$ for unreachable cells or infinite Bellman step cost from those cells.

C. Channel Geometric Guidance via Metric Tensor

The admissible set $U(x)$ determines which directions are feasible, but it does not fully describe how pedestrians approach and align with channel geometry before entering a passage. In real scenes, pedestrians often pre-align with the channel axis and merge smoothly near entrances rather than abruptly changing direction at the channel boundary. Therefore, after defining hard HJB constraints, we introduce a metric tensor $\mathbf{M}(x)$ to encode soft geometric preference within the feasible direction set.

To represent anisotropic channel geometries such as corridors and barriers, we introduce a spatially varying symmetric positive-definite metric tensor $\mathbf{M}(x)$ that generalizes the classical eikonal equation to anisotropic form [53]:

$$\sqrt{\nabla\phi^\top \mathbf{M}(x) \nabla\phi} = \frac{1}{f(\rho)}. \quad (9)$$

Under this metric, the optimal direction is determined by the tensor-induced steepest descent:

$$\mathbf{v} = f(\rho) \frac{-\mathbf{M}(x) \nabla\phi}{\sqrt{\nabla\phi^\top \mathbf{M}(x) \nabla\phi}}. \quad (10)$$

For a channel region Ω_c with unit tangent $\tau(x)$ and unit normal $n(x)$, we construct the metric tensor as:

$$\mathbf{M}(x) = \alpha(x)\tau\tau^\top + \beta(x)nn^\top, \quad x \in \Omega_c, \quad (11)$$

where $\alpha(x)$ and $\beta(x)$ control the traversal weights in tangential and normal directions, respectively. In the discrete Bellman update used in this paper, the directional step cost is proportional to $1/\sqrt{\mathbf{u}^\top \mathbf{M}(x) \mathbf{u}}$. Hence setting $\alpha(x) \gg \beta(x)$ lowers the effective cost along the channel tangent and suppresses cross-channel deviations, achieving continuous geometric guidance without explicit prohibition. Outside channels, $\mathbf{M}(x) = \mathbf{I}$ recovers the isotropic case.

Remark: The metric tensor represents directional preference rather than hard feasibility. As long as costs remain finite, pedestrians may still deviate when congestion or detours require it. Thus $U(x)$ answers which directions are allowed, whereas $\mathbf{M}(x)$ answers which allowed directions better follow the channel geometry.

D. Unified Discrete Bellman Formulation

To unify anisotropic geometry with discrete constraints, we employ a discrete Bellman formulation. For grid point x and admissible direction $\mathbf{u} \in U(x)$, define the step cost [56]:

$$\text{Cost}(x, \mathbf{u}) = \frac{\Delta x}{f(\rho(x))} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}(x) \mathbf{u}}}, \quad (12)$$

The potential satisfies:

$$\phi(x) = \min_{\mathbf{u} \in U(x)} [\phi(x + \Delta x \mathbf{u}) + \text{Cost}(x, \mathbf{u})]. \quad (13)$$

This formulation couples metric tensor $\mathbf{M}(x)$ for geometric guidance with control set $U(x)$ for directional constraints within a unified Bellman framework. The optimal direction is:

$$\mathbf{u}^*(x) = \arg \min_{\mathbf{u} \in U(x)} [\phi(x + \Delta x \mathbf{u}) + \text{Cost}(x, \mathbf{u})], \quad (14)$$

with velocity $\mathbf{v}(x) = f(\rho)\mathbf{u}^*(x)$.

E. Multi-Stage Multi-Route Model

Real-world scenarios often involve complex itineraries with multiple stages and route choices. We extend the density field to include stage and route indices, with fixed-probability splitting at decision boundaries.

1) *Stage-Route Decomposition*: Let the complete behavior chain consist of S stages, where stage s has R_s alternative routes. The total density decomposes as [57]:

$$\rho(x, t) = \sum_{s=1}^S \sum_{r=1}^{R_s} \rho_{s,r}(x, t), \quad (15)$$

where $\rho_{s,r}(x, t)$ is the density of sub-population in stage s following route r .

All sub-populations share the same speed function $f(\rho)$ determined by total density, ensuring collective congestion effects. Sub-populations differ in their potential functions, optimal directions, and exit targets.

2) *Sub-Population Potentials and Velocities*: For each sub-population (s, r) , define admissible direction set $U_{s,r}(x)$, metric tensor $\mathbf{M}_{s,r}(x)$, and target regions. The potential $\phi_{s,r}(x)$ satisfies:

$$\phi_{s,r}(x) = \min_{\mathbf{u} \in U_{s,r}(x)} \left[\phi_{s,r}(x + \Delta x \mathbf{u}) + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x) \mathbf{u}}} \right]. \quad (16)$$

The optimal direction is:

$$\mathbf{u}_{s,r}^*(x) = \arg \min_{\mathbf{u} \in U_{s,r}(x)} \left[\phi_{s,r}(x + \Delta x \mathbf{u}) + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x) \mathbf{u}}} \right], \quad (17)$$

with velocity $\mathbf{v}_{s,r}(x) = f(\rho) \mathbf{u}_{s,r}^*(x)$.

3) *Stage Transitions and Fixed-Probability Splitting*: Sub-populations couple through stage transitions. Let $G_{s,r} \subset \Omega$ be the completion goal region for stage (s, r) with indicator $\chi_{G_{s,r}}(x)$. Upon reaching this region, pedestrians switch to next-stage routes with fixed probabilities.

Let $p_{(s,r) \rightarrow q}$ be the fixed probability of switching from (s, r) to $(s+1, q)$, satisfying:

$$p_{(s,r) \rightarrow q} \geq 0, \quad \sum_{q=1}^{R_{s+1}} p_{(s,r) \rightarrow q} = 1. \quad (18)$$

The transition term is:

$$Q_{(s,r) \rightarrow (s+1,q)}(x, t) = p_{(s,r) \rightarrow q} \kappa_{s,r} \chi_{G_{s,r}}(x) \rho_{s,r}(x, t), \quad (19)$$

where $\kappa_{s,r} > 0$ is the stage switching rate.

The density evolution for each sub-population includes advection and transition source terms:

$$\frac{\partial \rho_{s,r}}{\partial t} + \nabla \cdot (\rho_{s,r} \mathbf{v}_{s,r}) = Q_{s,r}^{\text{in}} - Q_{s,r}^{\text{out}}, \quad (20)$$

where $Q_{s,r}^{\text{in}}$ is inflow from the previous stage and $Q_{s,r}^{\text{out}}$ is outflow to the next stage.

IV. NUMERICAL SOLUTION

We discretize the domain using a uniform two-dimensional grid, with the control direction set represented as an 8-neighborhood basis with masking for regional restrictions.

Unified Bellman Solver. Rather than separate eikonal and HJB solvers, we use a unified discrete Bellman shortest-path update. For candidate direction $\mathbf{u}$, precompute geometric step factors from the local tensor, then combine with the speed function to form the complete step cost. A priority queue propagates potential values outward from exits until convergence. When the control set is unrestricted, this procedure is equivalent to monotone solution of the anisotropic eikonal equation; when controls are restricted, it naturally reduces to constrained HJB solution, covering all required scenarios within a unified framework.

Density Update. After obtaining converged potentials, recover velocity fields using (17), substitute into the continuity equation, and advance density explicitly using an upwind flux scheme:

$$\rho^{n+1} = \rho^n - \Delta t \nabla \cdot (\rho^n \mathbf{v}^n). \quad (21)$$

The time step satisfies the CFL stability condition.

Evaluation Oracle. For any given control configuration $z = (s, \eta)$, the simulation workflow constitutes a lower-level evaluation operator:

$$(\rho, \phi, \mathbf{v}) = \mathcal{S}(z; \hat{p}), \quad (22)$$

where $\hat{p}$ represents fixed route preference parameters identified from historical data. This operator provides the foundation for subsequent control optimization.

V. CONTROL EVALUATION METRICS

We distinguish between behavior validation metrics (for verifying model mechanisms) and control evaluation metrics (for assessing policy impacts). This section focuses on the latter.

A. Efficiency Metric: Total Travel Time

For simulation horizon $[0, T]$, the total travel time metric is:

$$J_1(z | \hat{p}) = \int_0^T \int_{\Omega} \rho(x, t) dx dt. \quad (23)$$

This measures aggregate dwell time of the crowd, providing a stable basis for comparing different control strategies.

B. Safety Metric: High-Density Exposure Time

Let ρ_{safe} be the safe density threshold. The exposure time metric is:

$$J_2(z | \hat{p}) = \int_0^T \int_{\Omega} \mathbf{1}_{\rho(x,t) > \rho_{\text{safe}}} dx dt. \quad (24)$$

This captures cumulative space-time exposure to dangerous crowding conditions, reflecting management concerns about sustained congestion risks.

C. Balance Metric: Channel Flow Variance

Let $q_c(t)$ be the flux through channel c at time t , with cumulative flow $F_c = \int_0^T q_c(t) dt$. The load balancing metric is:

$$J_3(z | \hat{p}) = \text{Var}(F_1, \dots, F_m). \quad (25)$$

Lower variance indicates more even distribution of traffic across channels, reducing risk concentration and improving facility utilization.

D. Dimensionless Normalization

Because J_1 , J_2 , and J_3 have different physical units and numerical scales, we further define normalized indicators for policy comparison and optimization. Let $M_{\text{tot}} = M_{\text{init}} + M_{\text{in}}$ denote the total evaluated crowd mass, let $|\Omega_w|$ be the walkable area, and let $F = \sum_{c=1}^m F_c$ be the total cumulative channel flow. We define

$$\tilde{J}_1(z | \hat{p}) = \frac{J_1(z | \hat{p})}{M_{\text{tot}} T}, \quad (26)$$

$$\tilde{J}_2(z | \hat{p}) = \frac{J_2(z | \hat{p})}{|\Omega_w| T}, \quad (27)$$

and

$$\tilde{J}_3(z | \hat{p}) = \frac{\text{Var}(F_1, \dots, F_m)}{F^2(m-1)/m^2} = \frac{m^2}{m-1} \text{Var}(p_1, \dots, p_m), \quad (28)$$

where $p_c = F_c/F$. The normalized metrics are dimensionless, with $\tilde{J}_2 \in [0, 1]$ and $\tilde{J}_3 \in [0, 1]$.

E. Scalarized Optimization Objective

Given weights $(\lambda_1, \lambda_2, \lambda_3)$, the scalarized objective is:

$$J(z | \hat{p}) = \lambda_1 \tilde{J}_1(z | \hat{p}) + \lambda_2 \tilde{J}_2(z | \hat{p}) + \lambda_3 \tilde{J}_3(z | \hat{p}), \quad (29)$$

with the optimization problem:

$$\min_{z \in \mathcal{Z}} J(z | \hat{p}). \quad (30)$$

Our algorithm solves this scalarized problem for fixed weights, rather than directly generating the full Pareto frontier. Approximate Pareto sets can be obtained by running with different weight combinations.

VI. OPTIMIZATION METHOD

A. Control Variable Definition

We distinguish controllable variables from exogenous behavioral parameters:

Controllable variables:

- 1) *Channel direction configuration* $s = (s_1, \dots, s_C)$, where $s_c \in \{-1, 0, +1, \emptyset\}$ indicates reverse one-way, bidirectional, forward one-way, or closed state for channel c .
- 2) *Geometric guidance intensity* $\eta = (\eta_1, \dots, \eta_C)$, where $\eta_c = \alpha_c / \beta_c \geq 1$ is the anisotropy ratio.

Thus the optimization variable is $z = (s, \eta)$.

Exogenous parameters: Route preference parameters $\hat{p}$ are identified from historical data and fixed during optimization. Decision region locations are reserved for future extension.

B. Structure-Aware Hybrid Block Optimization

Treating this as a black-box optimization ignores three key structures: mixed discrete-continuous variables, the expensive multi-group simulation structure, and the different roles of discrete direction settings and continuous guidance intensities.

We propose Structure-Aware Hybrid Block Optimization (SA-HBO), with four core components:

- 1) Discrete block neighborhood search for channel directions;
- 2) Surrogate pre-screening to reduce full simulations;
- 3) Full simulation confirmation for retained candidates;
- 4) Continuous block projected stochastic approximation for guidance intensities.

C. Discrete Block Update

Fixing continuous variables η^k , the discrete subproblem at iteration k is:

$$s^{k+1/2} \approx \arg \min_{s \in \mathcal{S}} J(s, \eta^k | \hat{p}). \quad (31)$$

Define the neighborhood of current configuration s^k as:

$$\mathcal{N}(s^k) = \{s : \|s - s^k\|_0 \leq \nu\}, \quad (32)$$

where ν is the maximum number of simultaneously flipped channels (typically 1 or 2).

Surrogate pre-screening: To reduce full simulations, construct a local surrogate for each candidate $s' \in \mathcal{N}(s^k)$:

$$\hat{J}(s', \eta^k) = \mu_1 \hat{J}_1^{\text{local}} + \mu_2 \hat{J}_2^{\text{local}} + \mu_3 \hat{J}_3^{\text{flow}}, \quad (33)$$

where the local metrics approximate density changes, exposure changes, and flow redistribution. Select top K candidates by surrogate value for full PDE simulation.

D. Continuous Block Update

Fixing discrete variables $s^{k+1/2}$, solve for η :

$$\eta^{k+1} \approx \arg \min_{\eta \in \mathcal{H}} J(s^{k+1/2}, \eta | \hat{p}), \quad (34)$$

with feasible set $\mathcal{H} = \{\eta : \eta_c \geq 1\}$.

Since gradients are unavailable analytically, we use projected stochastic approximation. Generate Rademacher perturbation $\Delta_\eta^k \in \{-1, +1\}^C$, construct two-sided perturbations $\eta_\pm^k = \eta^k \pm c_k \Delta_\eta^k$, and project to feasible set. The stochastic gradient estimate is:

$$\hat{g}_\eta^k = \frac{J(s^{k+1/2}, \tilde{\eta}_+^k | \hat{p}) - J(s^{k+1/2}, \tilde{\eta}_-^k | \hat{p})}{2c_k} \odot (\Delta_\eta^k)^{-1}. \quad (35)$$

The candidate update is:

$$\eta_{\text{cand}}^{k+1} = \Pi_{\mathcal{H}}(\eta^k - \alpha_k D_\eta^{-1} \hat{g}_\eta^k), \quad (36)$$

with diagonal preconditioner D_η .

Acceptance/Rejection: Compare candidate objective J_{cand} with current J_{cur} . Accept if $J_{\text{cand}} \leq J_{\text{cur}} - \varepsilon_{\text{acc}}$; otherwise reject and optionally backtrack. Maintain best solution z_{best} throughout.

E. Algorithm Summary

- 1) Initialize potentials and density, compute $J(z^0 | \hat{p})$, set $z_{\text{best}} = z^0$.
- 2) For each iteration:
 - *Discrete update*: Generate neighborhood, screen with surrogate, evaluate top candidates with full simulation, select $s^{k+1/2}$.
 - *Continuous update*: Generate perturbations, compute stochastic gradient, generate candidate, accept/reject to obtain η^{k+1} .
 - *State update*: Set $z^{k+1} = (s^{k+1/2}, \eta^{k+1})$, compute objective, update z_{best} if improved.
- 3) Return $z^* = z_{\text{best}}$.

Stopping criteria include: small objective change, no discrete variable change for consecutive iterations, or maximum iteration/simulation/time limits.

VII. NUMERICAL EXPERIMENTS

This section presents numerical experiments to validate the proposed model mechanisms, control strategies, and optimization framework. The experiments use a four-channel scenario, which is a simplified abstraction of the Bund waterfront viewing-platform environment. One side of the platform contains the initial crowd region and entry boundary, the other side contains the viewing and exit region, and a central wall contains four representative passages: upper, middle, lower-middle, and lower passages. This abstraction preserves the essential spatial structure while reducing the actual passage system to four controllable channels, so that mechanism validation, strategy enumeration, and optimization search can be conducted in a finite and interpretable parameter space. The experiments follow a progressive logic: local mechanism validation is first conducted on the four-channel setting; the analysis then extends to multi-channel, multi-stage control strategies, behavior-layer validation, and finally joint optimization over channel direction and geometric guidance intensity.

A. Experimental Setup

1) *Four-Channel Scenario*: The four-channel scenic-platform scenario is used as the common basis for mechanism validation, strategy comparison, and optimization. The left side represents the initial crowd region and entry boundary, the right side represents the platform and exit target region, and the central wall contains four passages arranged from top to bottom as the upper, middle, lower-middle, and lower channels. Depending on the experiment group, pedestrians perform a single-stage crossing task, a bidirectional passage task, or a multi-stage “entry–tour–exit” behavior chain.

Numerical parameters are set as follows: grid resolution 128×96 , spatial step $\Delta x = 0.5$, free-flow speed $v_{\text{max}} = 1.5$, maximum density $\rho_{\text{max}} = 5.0$, initial density $\rho_{\text{init}} = 2.2$, CFL coefficient 0.9, and safety density threshold $\rho_{\text{safe}} = 3.5$. The mechanism validation experiment (G1) and the behavior-layer validation experiment (G3) use a time horizon of $T = 140$, while the control strategy comparison experiment (G2) and the optimal control search experiment (G4) use a multi-stage continuous-flow horizon of $T = 240$.

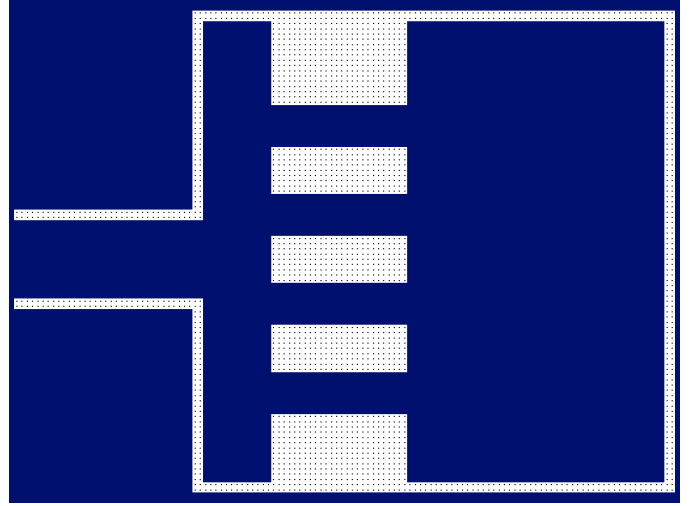


Fig. 2. Four-channel simulation scenario used in the numerical experiments.

TABLE I
G1 MECHANISM VALIDATION MATRIX

Mechanism Proposition	Comparison	Main Observables	Expected Phenomenon
$M(x)$ represents soft geometric guidance	baseline vs. M -only	channel flow share; approach streamlines	target-channel attraction expands; inflow converges earlier
$U(x)$ represents hard directional constraints	baseline vs. U -only	restricted-channel flow; local direction field	restricted direction is excluded; flow shifts to feasible channels
$U(x)$ suppresses reverse flow and counterflow	bidirectional baseline vs. one-way rule	reverse mass time; head-on cell time	reverse flow and counterflow intensity decrease
$U + M$ is spatially transferable	different controlled-channel locations	dominant flow channel; hotspot location	dominant flow migrates with the controlled channel

B. Mechanism Validation

1) *Validation Logic and Experimental Matrix*: The G1 mechanism validation experiment does not aim to rank control strategies. Instead, it verifies whether the two local mechanisms in the model have clear and interpretable behavioral meanings. The experiment is organized around four mechanism propositions, as shown in Table I.

This matrix separates the functional levels of $M(x)$ and $U(x)$. The former concerns preference changes within the feasible direction set, whereas the latter concerns modification of the feasible direction set itself. The final proposition then tests whether their combination produces synchronous migration when the controlled channel location changes.

2) *Soft Guidance Mechanism: $M(x)$ Alters Channel Attraction*: To verify whether geometric guidance changes the ap-

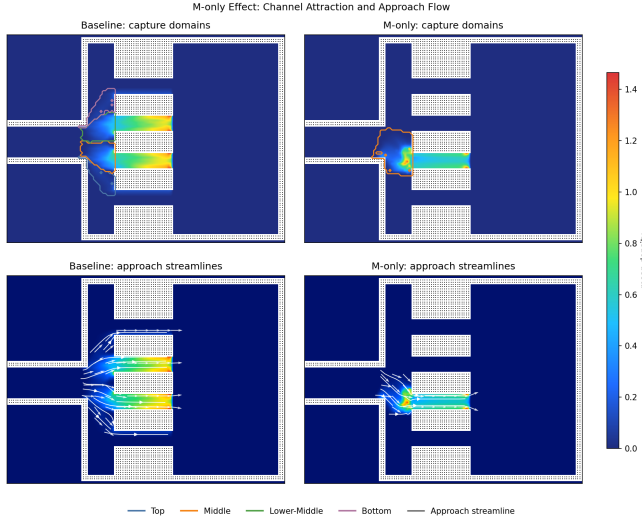


Fig. 3. Channel attraction domains and approach streamlines in the G1 experiment. The upper row shows capture-domain boundaries over the mean-density background, and the lower row shows time-averaged approach streamlines.

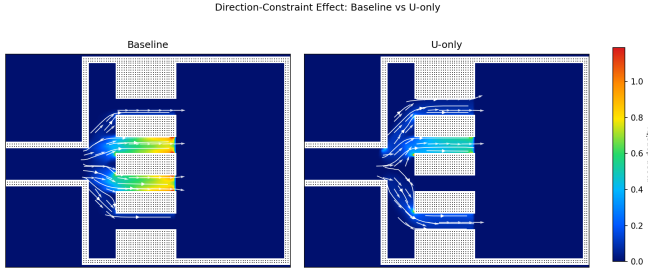


Fig. 4. Local direction field comparison between baseline (left) and U -only (right) configurations. White streamlines and arrows represent time-averaged movement directions; the background color encodes time-averaged density.

proach process, we compare the baseline configuration against the M -only configuration, in which anisotropic geometric guidance is applied to the middle channel.

The baseline flow is mainly distributed between the middle and lower-middle channels: upper 1.64%, middle 48.78%, lower-middle 47.92%, and lower 1.66%. After introducing M -only guidance, the middle-channel flow share increases to 100.00%. As shown in Fig. 3, M -only guidance not only expands the attraction domain of the middle channel but also causes the incoming flow to converge toward the middle channel further upstream, forming a clear pre-alignment structure before the channel entrance. This indicates that the core role of $M(x)$ is to alter the relative travel cost within the feasible direction set, reshaping which channel attracts the crowd and when alignment begins, rather than directly prohibiting other passages.

3) *Hard Directional Constraint Mechanism: $U(x)$* : To verify whether directional constraints alter feasible path selection, we first compare the baseline configuration against the U -only configuration under unidirectional inflow, where the middle channel is subject to a directional constraint.

Under the U -only configuration, the middle-channel flow

TABLE II
BIDIRECTIONAL $U(x)$ VALIDATION RESULTS

Configuration	Westbound Share	Counter-flow Mass	Head-on Exposure
Bidirectional baseline	73.18%	415.75	1213.76
Middle channel one-way	0	0	0

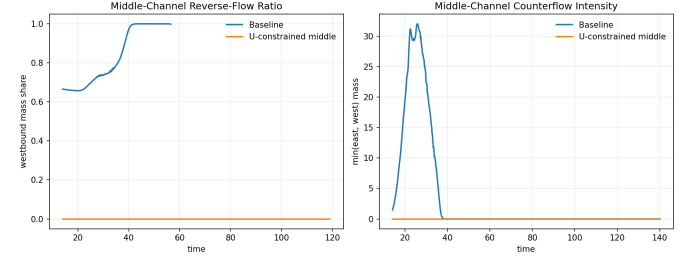


Fig. 5. Dynamic comparison of reverse flow in the middle channel under bidirectional conditions. Left: westbound mass share in the middle channel over time. Right: head-on conflict intensity in the middle channel over time.

share drops to 0. The flow is redistributed to the upper, lower-middle, and lower channels, with shares of 19.25%, 68.76%, and 12.00%, respectively. This is the primary evidence for the role of $U(x)$: directional constraints do not continuously adjust route preference, but directly modify the local admissible direction set and thereby remove the restricted channel from the feasible choice set.

To further verify the ability of $U(x)$ to control directional conflicts, we conduct a bidirectional flow sub-experiment comparing a bidirectional baseline against a middle-channel eastbound one-way rule.

The bidirectional experiment shows that after imposing the one-way rule, westbound pedestrians are completely excluded from the middle channel, and both the middle-channel reverse-flow mass time and head-on cell time become zero. Therefore, $U(x)$ does not merely change path choice under unidirectional demand; it also persistently suppresses reverse flow and head-on conflict under bidirectional demand. This distinction clarifies the functional difference between $U(x)$ and $M(x)$: $M(x)$ adjusts route preference, whereas $U(x)$ modifies the feasible direction set. Thus, $U(x)$ is the hard-constraint representation of directional control policies.

4) *Composability and Spatial Transferability of the Joint Mechanism*: To verify the sensitivity of the joint mechanism to controlled-channel location, we test four $U + M$ configurations: middle-channel controlled, upper-channel controlled, lower-middle-channel controlled, and lower-channel controlled. The focus here is not on the efficiency or safety index of each configuration, but on whether the dominant flow channel changes synchronously when $U + M$ is applied to different channels.

The results show that changing the controlled-channel location induces synchronous migration of the dominant flow channel, density hotspot, and approach streamlines. When the upper, middle, lower-middle, or lower channel is selected as the joint guidance channel, the flow concentrates on the corresponding channel. This confirms that $U + M$ is not merely a local parameter setting that works at one fixed position; rather, it produces clear and interpretable behavioral responses

TABLE III
DOMINANT FLOW MIGRATION UNDER DIFFERENT $U + M$
CONTROLLED-CHANNEL LOCATIONS

Configuration	Dominant Flow Channel	Upper Flow	Middle Flow	Lower-Middle Flow	Lower Flow
baseline	middle/lower-middle	1.64%	48.78%	47.92%	1.66%
M -only-middle	middle	0	100.00%	0	0
$U + M$ -middle	middle	0	100.00%	0	0
$U + M$ -top	upper	100.00%	0	0	0
$U + M$ -lower-middle	lower-middle	0	0	100.00%	0
$U + M$ -lower	lower	0	0	0	100.00%

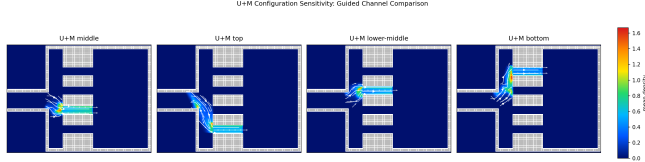


Fig. 6. Flow field comparison for the $U + M$ combined mechanism under different controlled-channel locations. The sub-figures correspond to different guided channels; white streamlines and arrows represent time-averaged movement directions, and the background color encodes time-averaged density.

as the spatial configuration changes. The joint mechanism can therefore serve as the lower-level simulator for subsequent strategy evaluation and optimization search.

C. System Response and Optimization Complexity Under Directional Control

After validating that $U(x)$ and $M(x)$ produce the expected local control responses, we further examine whether the discrete directional control variable s induces nontrivial system-level responses in the full multi-stage scenario. The purpose of this experiment is not to select a final optimal policy from a small enumeration, but to test whether interpretable perturbations of channel directions generate distinguishable changes in efficiency, safety exposure, and load balancing. Such evidence is needed before introducing an automated optimization procedure.

The complete combinatorial space of four-channel direction settings is large and contains many cases with weak operational meaning. We therefore construct a representative validation matrix with three policy families: single-entry (SE), single-return (SR), and one-channel-closed (CL), each rotated over the upper, middle, lower-middle, and bottom channels. Together with the all-free baseline (BSL), this gives $1 + 4 + 4 + 4 = 13$ cases. Direction codes are ordered as $[T, M, LM, B]$, where T , M , LM , and B denote the upper, middle, lower-middle, and bottom channels. The symbols E , W , F , and X denote eastbound entry, westbound return, free bidirectional passage, and closure, respectively. Case indices are retained only as configuration identifiers and are not mixed with policy-family labels.

Fig. 7 shows that different direction settings produce stable and clearly separated changes in $\tilde{J}_1$, $\tilde{J}_2$, and $\tilde{J}_5$. Using the aggregated normalized objective $\tilde{J} = \tilde{J}_1 + \tilde{J}_2 + \tilde{J}_5$, C8 (SR-LM, EEWE) obtains the lowest value, $\tilde{J} = 0.5102$. The baseline C1 is close, with $\tilde{J} = 0.5145$, but achieves the lowest safety exposure, $\tilde{J}_2 = 0.00785$. C9 (SR-B) gives the best load-balancing term, $\tilde{J}_5 = 0.1138$, while C3 (SE-M) gives the best

efficiency term, $\tilde{J}_1 = 0.3141$, at the cost of higher safety exposure and load imbalance.

These results indicate that perturbing the direction variable s does not produce a monotonic improvement pattern, nor does any single configuration dominate all three objectives. The non-dominated set contains C1, C3, C4, C6, C8, and C9, confirming that the four-channel multi-stage scenario already forms a nontrivial multi-objective response structure.

The direct spatial mechanism behind these objective differences is the redistribution of channel loads and high-density hotspots. As shown in Fig. 8(A), the baseline C1 concentrates flow mainly in the middle and lower-middle channels, with shares of 34.0% and 47.2%, respectively. C8 has channel-load shares of 0.6%, 47.6%, 32.0%, and 19.7%, maintaining a low aggregate objective while still assigning the largest burden to the middle channel. C9 has load shares of 0, 30.0%, 33.7%, and 36.3%, which explains its best load-balancing term but weaker efficiency.

Fig. 8(B) further shows that directional control changes not only total flow allocation but also the spatial location of risk exposure. Hotspot locations are visualized in the true scene coordinate system and on the same feasible-domain and obstacle background as the density maps. Upper-channel-dominant or middle-closure configurations shift hotspots upward; middle-channel dominance places hotspots near the central passage; and lower-middle or bottom involvement shifts hotspots downward. Hence, the variations in $\tilde{J}_1$, $\tilde{J}_2$, and $\tilde{J}_5$ are not merely abstract numerical differences, but arise from coupled changes in channel-load redistribution, local safety exposure, and hotspot migration.

The G2 experiment therefore supports the need for subsequent optimization from three perspectives. First, the problem is intrinsically multi-objective: efficiency, safety, and load balancing do not share a consistent optimum. Second, the direction variable is combinatorial, and the 13 cases here are only an interpretable slice rather than an exhaustive enumeration. Third, the full control problem must also optimize the continuous geometric guidance strength η . The resulting problem is therefore a mixed discrete-continuous, simulation-driven optimization problem with expensive evaluations, motivating the automated search procedure introduced later.

D. Multi-Stage Flow Validation Experiments

This subsection validates the multi-stage behavior chain and probabilistic splitting mechanism. The experiments compare single-stage approximations against full multi-stage models to demonstrate how stage transitions and fixed-probability splitting alter density evolution and exit load distribution.

Detailed results and analysis will be presented in the full version of this paper.

E. Optimal Control Parameter Search

This subsection presents the first-version automated parameter search results. The search enumerates discrete strategy variables, performs coarse grid search on guidance intensity η , and searches over fixed splitting probability candidates to

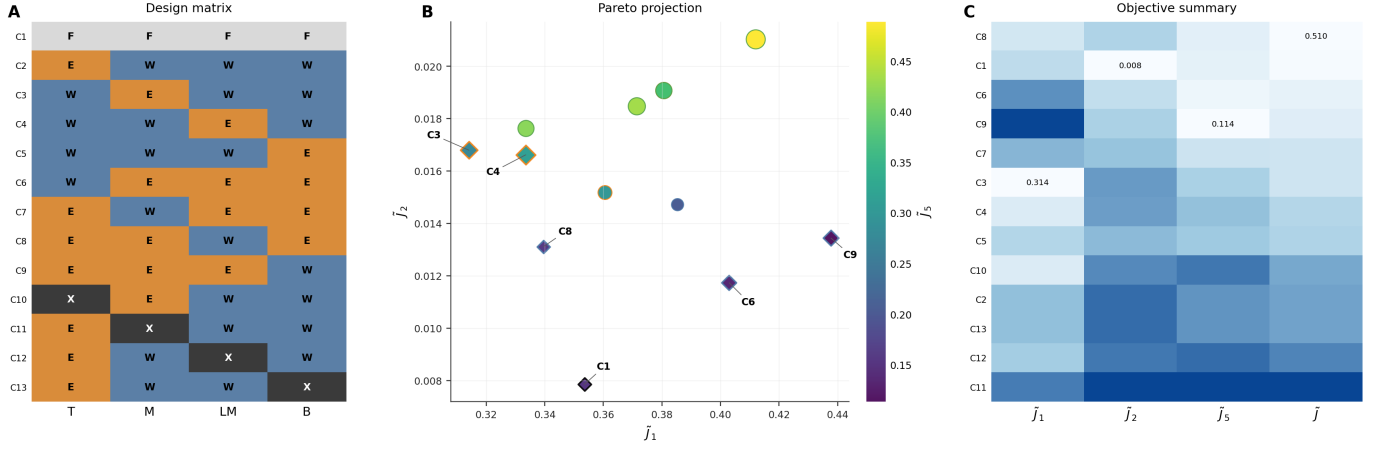


Fig. 7. Control-variable design and objective-space trade-off in G2. (A) Directional design matrix ordered by $[T, M, LM, B]$. (B) Pareto projection, where color encodes $\tilde{J}_5$, marker size encodes the aggregated normalized objective $\tilde{J}$, and diamond markers indicate non-dominated solutions. (C) Objective summary heatmap, with only the best value in each objective dimension annotated.

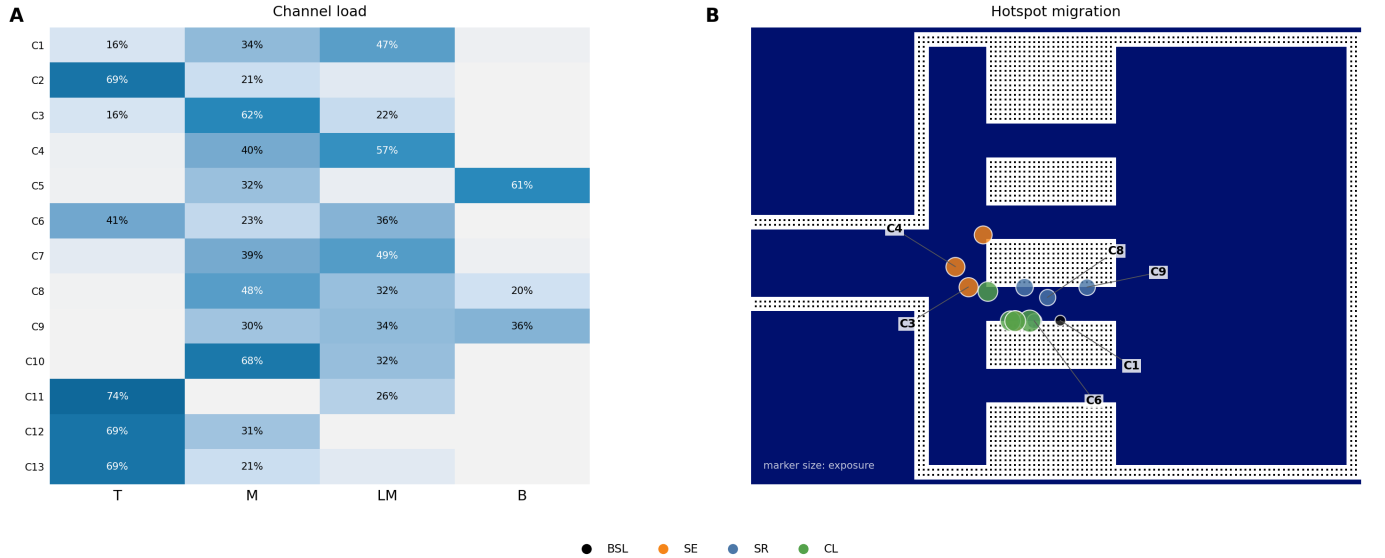


Fig. 8. Spatial mechanism behind directional-control responses. (A) Channel-load distribution across the 13 representative cases, with annotations retained only for major loaded channels. (B) Hotspot migration over the same scene background used in the density maps. Marker size represents local exposure intensity, and color indicates the policy family. If an exposure-weighted centroid falls inside an obstacle due to averaging over separated high-density regions, it is mapped to the nearest walkable cell for visualization.

TABLE IV
NON-DOMINATED DIRECTIONAL-CONTROL CASES IN G2

Case	Policy	Code	$\tilde{J}_1$	$\tilde{J}_2$	$\tilde{J}_5$	$\tilde{J}$	Main characteristic
C1	BSL	FFFF	0.3535	0.00785	0.1531	0.5145	Lowest exposure; close to the best aggregate objective
C3	SE-M	WEWW	0.3141	0.01680	0.2741	0.6049	Best efficiency; risk concentrates near the middle channel
C4	SE-LM	WWEW	0.3335	0.01662	0.3131	0.6632	Lower-middle entry; efficient but load-imbalanced
C6	SR-T	WEEE	0.4029	0.01174	0.1335	0.5482	Highest exit throughput; relatively balanced
C8	SR-LM	EEWE	0.3395	0.01311	0.1576	0.5102	Lowest aggregate objective
C9	SR-B	EEEW	0.4377	0.01344	0.1138	0.5649	Most balanced channel loads

identify optimal control configurations under different objective weight combinations $(\lambda_1, \lambda_2, \lambda_3)$.

Detailed results and analysis will be presented in the full version of this paper.

VIII. CONCLUSION

This paper presents a bi-level optimization framework for crowd channel management that combines discrete directional constraints with anisotropic geometric guidance. The proposed SA-HBO algorithm exploits problem structure through surrogate pre-screening, full simulation confirmation, and block

coordinate updates to handle the expensive simulation-based optimization efficiently.

The numerical experiments validate that: (1) geometric guidance $M(x)$ and directional constraints $U(x)$ authentically alter local crowd behaviors; (2) the combined model exhibits stable sensitivity to configuration changes; (3) directional constraints effectively suppress counter-flow and head-on conflicts under bidirectional demand. These findings establish the foundation for subsequent strategy optimization studies.

Future work will extend to full multi-objective optimization, sensitivity analysis under demand perturbations, and calibration against real-world observation data.

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